BUBBLE UI · DESIGN BOOK

Five landscapes. One interface.

ஐந்திணை

Sangam poetry sorts the world into five tinai, each a landscape with its own mood. The Bubble UI gives each one a whole environment: a ground and an ink, a way of moving, a living background and a figure that answers your hand. A Yazhi app should feel like where it belongs before a word is read.

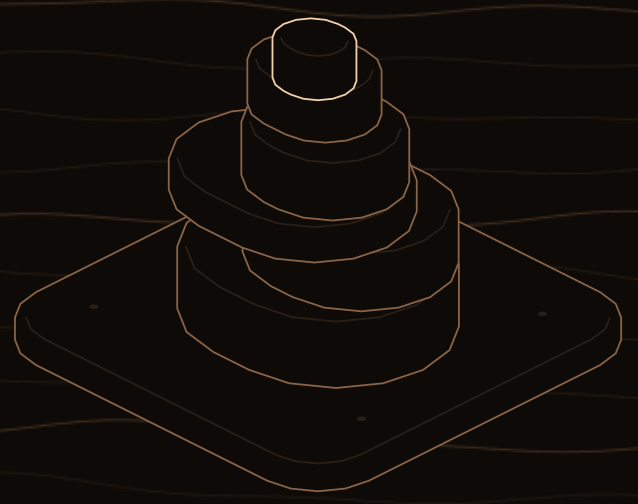
பாலை · Palai | முல்லை · Mullai | குறிஞ்சி · Kurinji | மருதம் · Marutham | நெய்தல் · Neytal

Heat over sand. Dune contours bend away from the pointer like heat haze.

அ நி கு கு க ச யா

Avai

rest



gap

Move over the figure. Hairline figures after Hairline by Lucas Marques (MIT).

01 Principles 02 The tinai system 03 Palai 04 Mullai 05 Kurinji 06 Marutham 07 Neytal 08 Karupporul atlas 09 Sangam objects 10 Foundations 11 Motion 12 Access and Tamil 13 Do and don't

01

Principles · அறம்

The interface is where Yazhi's Aram rules become visible. Each rule below has a component that enforces it, so following the design book is following the rules.

No claim without a source

சான்றின்றி சொல்லேல்

Every answer carries its citations inline. If there is no source, the bubble says "I don't know" and offers where to look.

Data stays home

தரவு வீட்டிலேயே

A runtime badge always shows where the model is running. On-device is the default; the cloud is a visible choice.

Think, then act

சிந்தித்து செயல்

Nothing changes a system without a confirm gate that names exactly what will happen, in words, before it happens.

Children first

குழந்தைகள் முதலில்

Child-facing screens are held until counsel signs off. Parents see everything a child's bubble does.

Ship what we said

சொன்னதைச் செய்

Drafts are marked as drafts — including Tamil copy awaiting a native reviewer. Nothing pretends to be finished.

Tamil is first-class

தமிழ் முதன்மை

Tamil gets its own line height, is never letter-spaced and is never cut mid-letter. Labels are bilingual by default.

Tamil phrasing above is a draft pending native review.

The hairline doctrine

Our figures and motion follow the ten rules of Hairline by Lucas Marques: isometric line drawings that answer the pointer. We vendor its MIT-licensed kernel

01 Hit areas never move: test against the rest pose.

02 Stagger by distance from where the change started.

and draw our own figures, one per landscape, to its rules. The same rules govern every animated surface in the Bubble UI.

03 Clamp reach; let falloff keep a floor.

04 Stroke is the only highlight: no fills, no glows.

05 Design the rest pose; one bright mark.

06 Honest construction, painted back to front.

07 Loops sleep offscreen.

08 Two clocks: a tween for choices, springs for the pointer.

09 Rounded solids are a silhouette and one crease.

10 No words in the figure; the read-out names the part.

02

The tinai system · திணை

Every environment keeps the brand's spine — an ink ground, cream text, one accent — and changes the temperature of the ink, the accent hue and the way things move. Wrap any subtree in `<TinaiScope tinai="…">` and everything inside takes the landscape; change it and every surface and stroke cross-fades.

TINAI	SIGNATURE	MOTION	USE FOR	APPS
பாலை Palai	Heat over sand	Heavy and unhurried.	Security, infrastructure and hard engineering	Kural Desk, yazhi-api console
முல்லை Mullai	Fireflies under the canopy	Gentle and patient.	Community, Dev Spaces and Yazhi Academy	Guru, Dev Spaces
குறிஞ்சி Kurinji	Mist on the ridge	Buoyant.	Launches and first releases	Avai, onboarding
மருதம் Marutham	Wind over paddy	Precise and fair.	Civic and legal: Nyaya, Sezhi; everyday commerce	Nyaya, Kadai
நெய்தல் Neytal	Tide lines	Tidal.	Diaspora, Yazh, and language loss	Open Sangam, Yazh (parents)

03 · DRYLANDS

பாலை

Palai

Heat over sand.

Endurance — courage and separation, the harder stories.

LIGHT

Low sun from the west; warm ink, contours instead of surfaces.

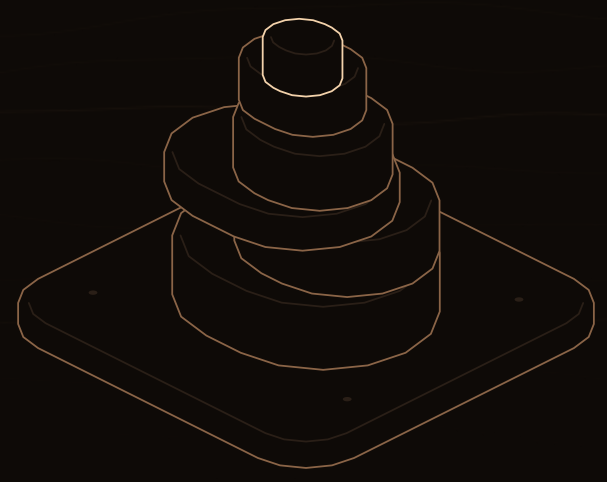
POINTER

Dune contours bend away from the pointer like heat haze.

USE FOR

Security, infrastructure and hard engineering. Default home of Kural Desk and yazhi-api console.

rest



gap

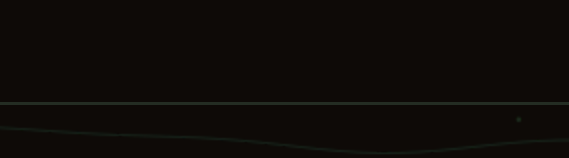
PALETTE · CLICK TO COPY

IN THE BUBBLE UI

ground	surface	hi	edge	mid	lo	accent	text	text-dim
#0E0A07	#18110C	#F3D2AB	#8A6447	#4D3828	#2A1F17	#C25B3C	#F6ECE0	#C9B8A6

MOTION PERSONALITY

Heavy and unhurried. Things travel far and settle late.



Send it

settle 900ms · ambient pace ×0.35

k 70 c 17 ζ 1.02 overshoot 0%




```
<TinaiScope tinai="palai">
  <TinaiAmbient tinai="palai" />
  <AppFrame app={…} />
</TinaiScope>
```

KARUPPORUL · கருப்பொருள்

the full atlas

BIRD	BEAST	FLOWER
புறா · பருந்து dove, kite	செந்நாய் · வலியிழந்த யானை wild dog, worn-out elephant	குரவம் · பாதிரி kuravam, trumpet flower
TREE	**DRUM**	**YAZH**
இலுப்பை · பாலை mahua, palai tree	துடி tudi, the hourglass drum	பாலை யாழ் palai yazh

Midday, summer, late dew; the feeling is separation (பிரிதல்).

04 · FOREST

முல்லை

Mullai

Fireflies under the canopy.

Waiting — patience, animals and the folk tales children know.

LIGHT

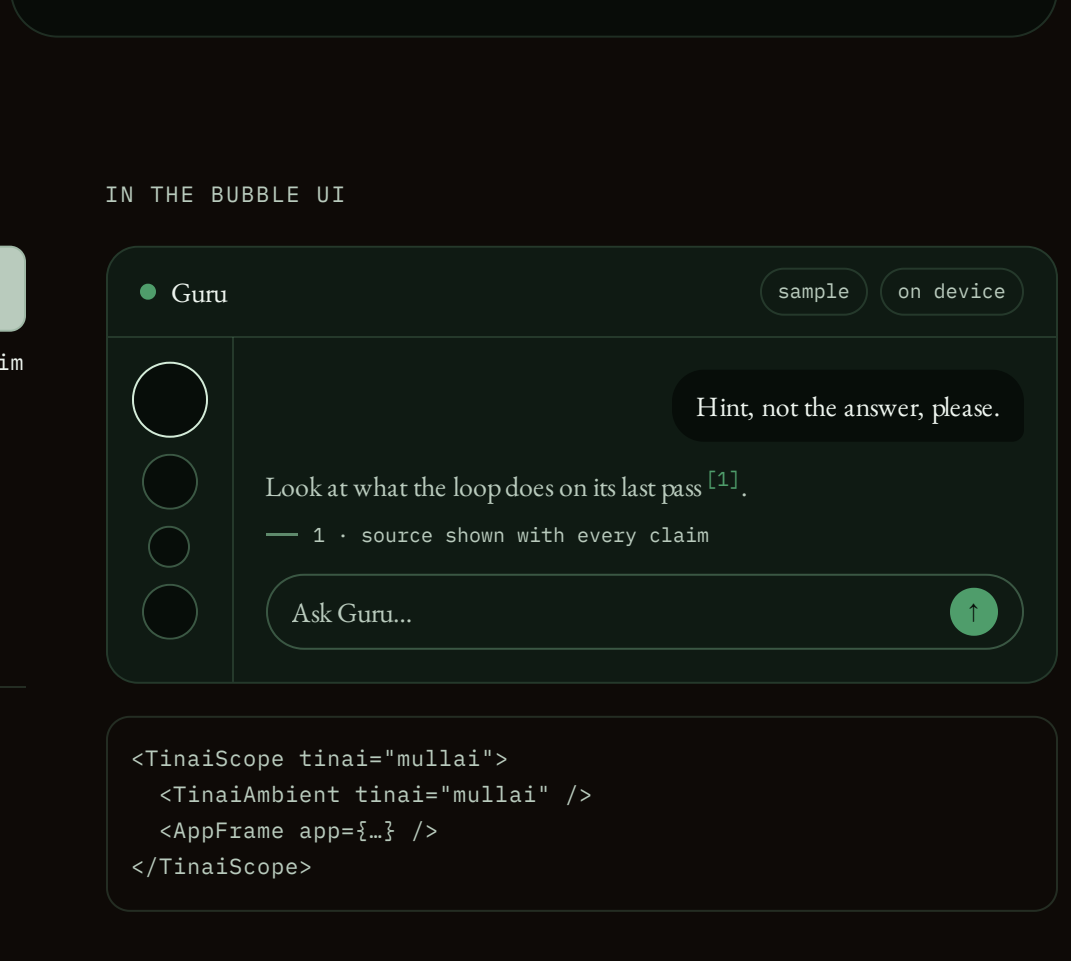
Dappled; green-black ink, small lights that come and go.

POINTER

Fireflies drift toward the pointer, slowly, and keep their distance.

USE FOR

Community, Dev Spaces and Yazhi Academy. Default home of Guru and Dev Spaces.

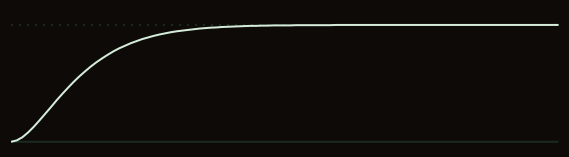


PALETTE · CLICK TO COPY

ground	surface	hi	edge	mid	lo	accent	text	text-dim
#070D09	#0F1A13	#D4ECD9	#5F8A6C	#2F4A38	#1A2A20	#4F9D6B	#EEF5EF	#B9CBBD

MOTION PERSONALITY

Gentle and patient. Nothing hurries; everything arrives.



Send it

settle 760ms · ambient pace ×0.5

k 85 c 18 ζ 0.98 overshoot 0%

IN THE BUBBLE UI


```
<TinaiScope tinai="mullai">
  <TinaiAmbient tinai="mullai" />
  <AppFrame app={…} />
</TinaiScope>
```

KARUPPORUL · கருப்பொருள்

the full atlas

BIRD காட்டுக்கோழி jungle fowl	BEAST மான் · முயல் deer, hare	FLOWER முல்லை · தோன்றி jasmine, glory lily
TREE கொன்றை · காயா · குருந்தம் golden shower, kaya, wild lime	DRUM ஏறுகோட் பறை bull-taming drum	YAZH முல்லை யாழ் mullai yazh

Evening, rains; the feeling is patient waiting (இருத்தல்).

05 · MOUNTAINS

குறிஞ்சி

Kurinji

Mist on the ridge.

First meetings — curiosity and discovery, where a story starts.

LIGHT

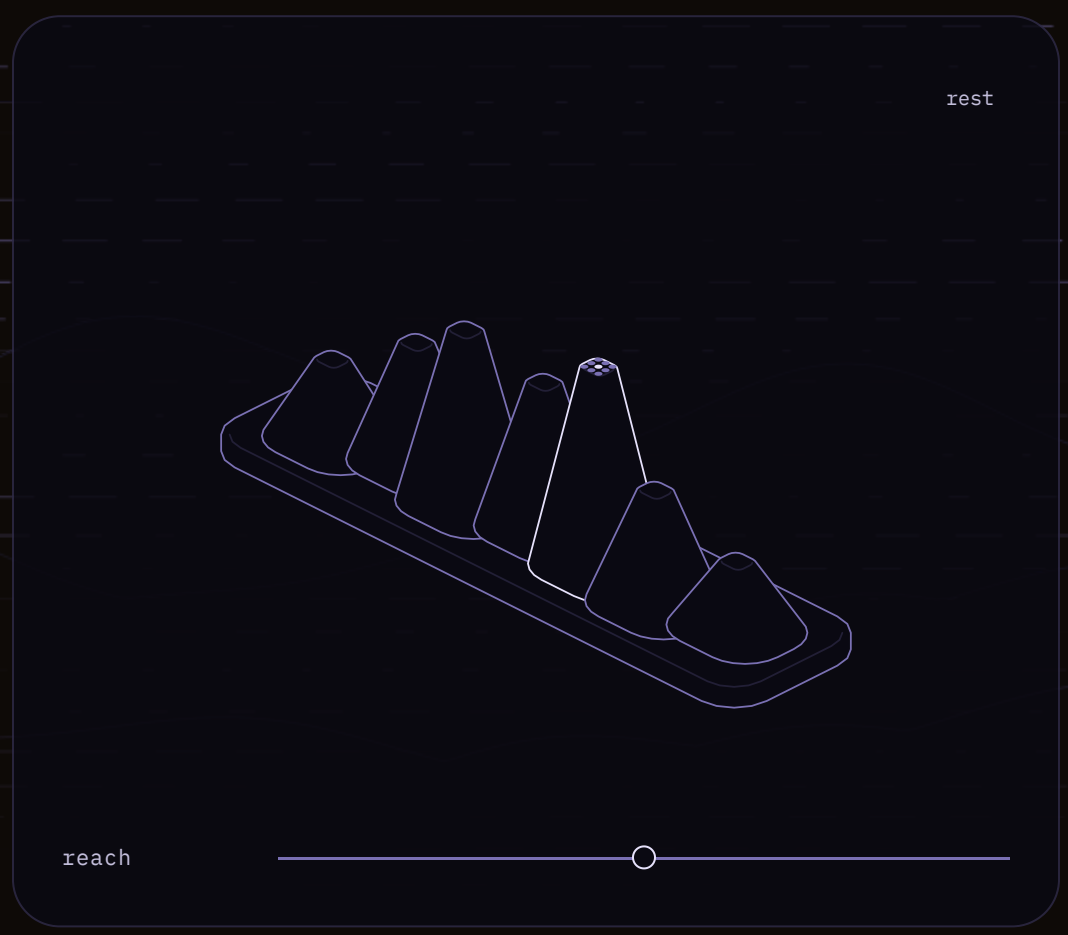
High, cool and clear; violet ink, ridgelines layered into distance.

POINTER

The mist parts where the pointer passes and closes behind it.

USE FOR

Launches and first releases. Default home of Avai and onboarding.



PALETTE · CLICK TO COPY

ground	surface	hi	edge	mid	lo	accent	text	text-dim
#0A0912	#14121F	#E6E1FB	#7A70B3	#3D3760	#221F36	#8B7AE0	#F2F0FB	#C4BFDC

MOTION PERSONALITY

Buoyant. A little lift, a small overshoot, then still.



k 120 c 15 ζ 0.68 overshoot 5%

Send it

settle 700ms · ambient pace ×0.7

IN THE BUBBLE UI

Avai

sample · on device

Where do I start?

Pick one landscape and build one bubble [1].

— 1 · source shown with every claim

Ask Avai...

```
<TinaiScope tinai="kurinji">
  <TinaiAmbient tinai="kurinji" />
  <AppFrame app={…} />
</TinaiScope>
```

KARUPPORUL · கருப்பொருள்

the full atlas

BIRD கிளி · மயில் parrot, peacock	BEAST புலி · யானை · கரடி tiger, elephant, bear	FLOWER குறிஞ்சி · காந்தள் kurinji, glory lily
TREE சந்தனம் · வேங்கை · மூங்கில் sandalwood, venkai, bamboo	DRUM தொண்டகப் பறை tondagam drum	YAZH குறிஞ்சி யாழ் kurinji yazh

Midnight, autumn, early dew; the feeling is union (புணர்தல்).

06 · FARMLAND

மருதம்

Marutham

Wind over paddy.

Everyday life — counting, work and family; quarrel and reconciliation.

LIGHT

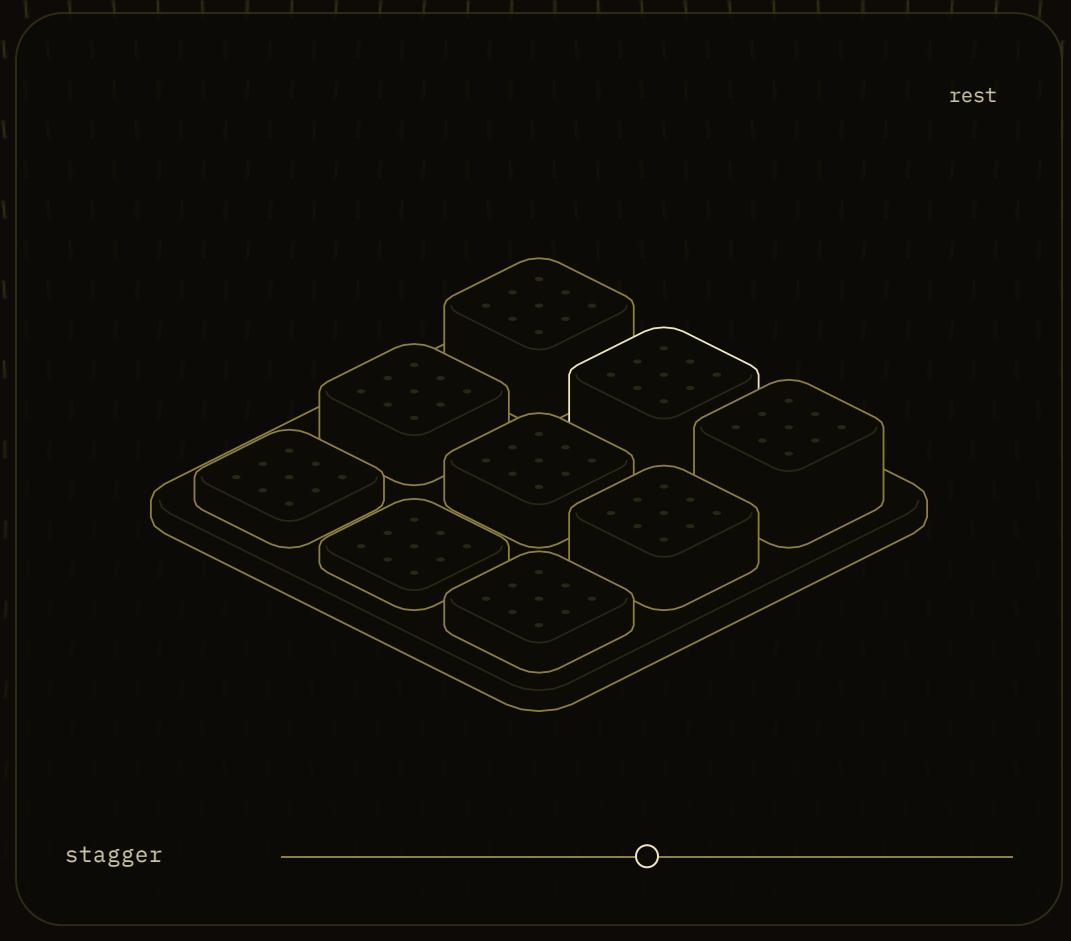
Even daylight; olive ink, rows and plots, nothing hidden.

POINTER

The pointer is the wind; seedlings bow in a wave that spreads from it.

USE FOR

Civic and legal: Nyaya, Sezhi; everyday commerce. Default home of Nyaya and Kadai.



PALETTE · CLICK TO COPY

ground	surface	hi	edge	mid	lo	accent	text	text-dim
#0C0B06	#17150C	#EFE6BD	#8E7F3E	#4A4222	#2A2614	#B7A03C	#F7F3E3	#CDC6A6

MOTION PERSONALITY

Precise and fair. Critically damped: it arrives exactly, never bounces.

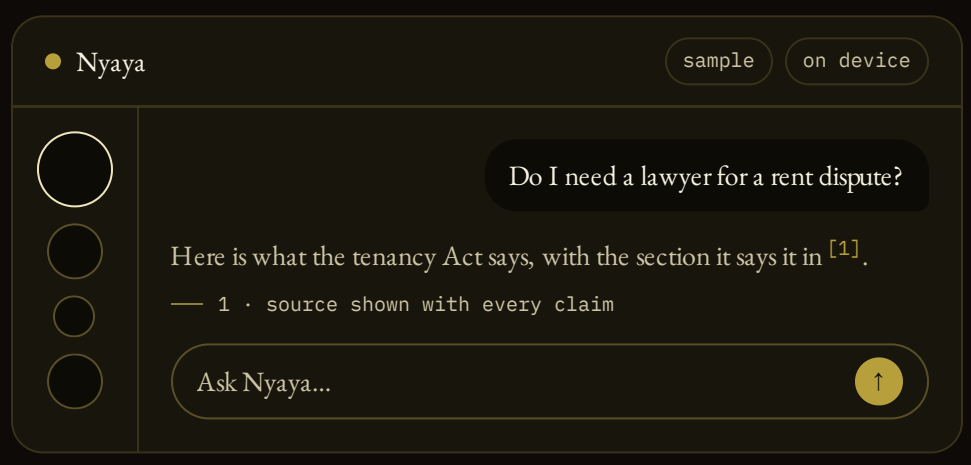


k 100 c 20 ζ 1.00 overshoot 0%

Send it

settle 700ms · ambient pace ×0.6

IN THE BUBBLE UI




```
<TinaiScope tinai="marutham">
  <TinaiAmbient tinai="marutham" />
  <AppFrame app={…} />
</TinaiScope>
```

KARUPPORUL · கருப்பொருள்

the full atlas

BIRD	BEAST	FLOWER
நாரை · குருகு stork, heron	எருமை · நீர்நாய் buffalo, otter	தாமரை · கழுநீர் lotus, blue water lily
TREE	**DRUM**	**YAZH**
மருதம் · காஞ்சி arjuna, kanji	நெல்லரி கிணை · மண முழவு harvest drum, wedding drum	மருத யாழ் marutha yazh

Before dawn, every season; the feeling is lovers' quarrel (உளறல்).

07 · COAST

நெய்தல்

Neytal

Tide lines.

Longing — voyages and distance, the diaspora's landscape.

LIGHT

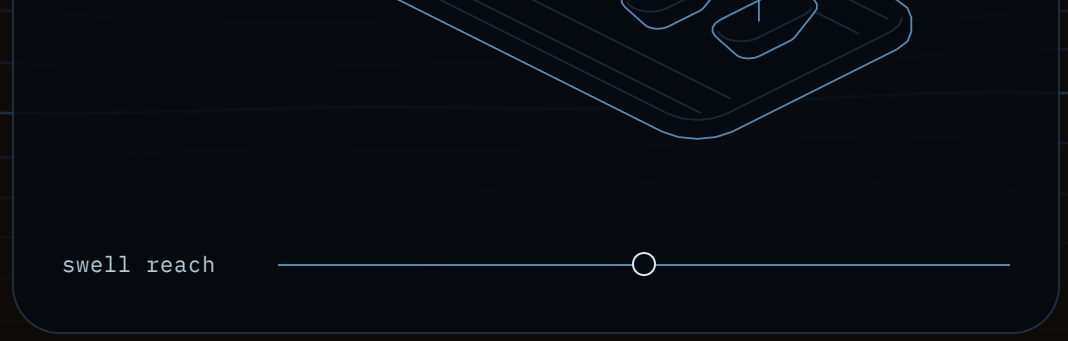
Moonlit; blue-black ink, long horizontal lines that breathe.

POINTER

The pointer drops a ripple; rings spread and fade into the tide.

USE FOR

Diaspora, Yazh, and language loss. Default home of Open Sangam and Yazh (parents).



PALETTE · CLICK TO COPY

ground	surface	hi	edge	mid	lo	accent	text	text-dim
#050A11	#0C1623	#D6E8F6	#5A87AC	#2B4762	#172738	#4A8AB5	#EDF4FA	#B7C8D6

MOTION PERSONALITY

Tidal. Long follow-through, a slow sway before it rests.

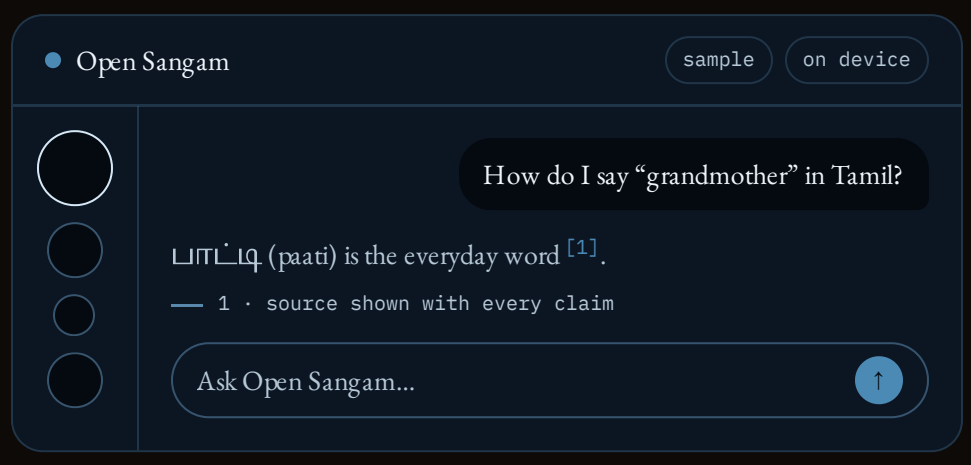


k 55 c 10 ζ 0.67 overshoot 6%

Send it

settle 850ms · ambient pace ×0.45

IN THE BUBBLE UI




```
<TinaiScope tinai="neytal">
  <TinaiAmbient tinai="neytal" />
  <AppFrame app={…} />
</TinaiScope>
```

KARUPPORUL · கருப்பொருள்

the full atlas

BIRD	BEAST	FLOWER
கடற்காக்கை sea crow	சுறா · முதலை shark, crocodile	நெய்தல் · தாழை water lily, screwpine
TREE	**DRUM**	**YAZH**
புன்னை · கண்டல் punnai, mangrove	மீன்கோட் பறை fish-catch drum	விளரி யாழ் vilari yazh

Late afternoon, every season; the feeling is lament (இரங்கல்).

08

Karupporul atlas · கருப்பொருள்

The things native to each landscape — its god, birds and beasts, flowers and trees, food, water, drum, yazh and melody, work and settlement. These are the raw material for each environment's icons, sounds and illustrations. Hover a column to light it; click a landscape to go to its chapter.

Draft — compiled from Tolkāppiyam and its standard teaching summaries; awaiting a Tamil scholar's review.

Both | தமிழ் | English

கரு · KARU	பாலை Palai	முல்லை Mullai	குறிஞ்சி Kurinji	மருதம் Marutham	நெய்தல் Neytal
நிலம் LAND	சுரம் (வறண்ட நிலம்) parched wasteland	காடு forest and pasture	மலை mountains	வயல் farmland	கடல் seashore
பெரும்பொழுது SEASON	வேனில், பின்பனி summer, late dew	கார் rains	கூதிர், முன்பனி autumn, early dew	எல்லாப் பருவமும் every season	எல்லாப் பருவமும் every season
சிறுபொழுது HOUR	நண்பகல் midday	மாலை evening	யாமம் midnight	வைகறை before dawn	எற்பாடு late afternoon
உரிப்பொருள் FEELING	பிரிதல் separation	இருத்தல் patient waiting	புணர்தல் union	ஊடல் lovers' quarrel	இரங்கல் lament
தெய்வம் DEITY	கொற்றவை Korravai	மாயோன் Mayon	சேயோன் (முருகன்) Seyon (Murugan)	வேந்தன் (இந்திரன்) Vendan (Indra)	வருணன் (கடலோன்) Varunan

புள் BIRD	புறா dove	காட்டுக்கோழி jungle fowl	கிளி parrot	நாரை stork	கடற்காக்கை sea crow
	பருந்து kite		மயில் peacock	குருகு heron	
மா BEAST	செந்நாய் wild dog	மான் deer	புலி tiger	எருமை buffalo	சுறா shark
	வலியிழந்த யானை worn-out elephant	முயல் hare	யானை elephant	நீர்நாய் otter	முதலை crocodile
			கரடி bear		
பூ FLOWER	குரவம் kuravam	முல்லை jasmine	குறிஞ்சி kurinji	தாமரை lotus	நெய்தல் water lily
	பாதிரி trumpet flower	தோன்றி glory lily	காந்தள் glory lily	கழுநீர் blue water lily	தாழை screwpine
மரம் TREE	இலுப்பை mahua	கொன்றை golden shower	சந்தனம் sandalwood	மருதம் arjuna	புன்னை punnai
	பாலை palai tree	காயா kaya	வேங்கை venkai	காஞ்சி kanji	கண்டல் mangrove
		குருந்தம் wild lime	மூங்கில் bamboo		
உணா FOOD	வழிப்பறிப் பொருள் spoils of the road	வரகு kodo millet	தினை foxtail millet	செந்நெல் red paddy	மீன் fish
		சாமை little millet	மூங்கிலரிசி bamboo rice	வெண்ணெல் white paddy	உப்பு விற்ற பண்டம் goods bartered for salt
நீர் WATER	வற்றிய சுனை dried spring	காட்டாறு forest stream	அருவி waterfall	ஆறு river	கடல் sea
	கூவல் dug pit		சுனை mountain spring	பொய்கை pond	உவர்க்குழி brackish pit
பறை DRUM	துடி tudi, the hourglass drum	ஏறுகோட் பறை bull-taming drum	தொண்டகப் பறை tondagam drum	நெல்லரி கிணை harvest drum	மீன்கோட் பறை fish-catch drum
				மண முழவு wedding drum	
யாழ் YAZH	பாலை யாழ் palai yazh	முல்லை யாழ் mullai yazh	குறிஞ்சி யாழ் kurinji yazh	மருத யாழ் marutha yazh	விளரி யாழ் vilari yazh
பண் MELODY	பஞ்சுரம் panchuram	சாதாரி sadari	குறிஞ்சிப் பண் kurinji pann	மருதப் பண் marutham pann	செவ்வழி sevvazhi
தொழில் WORK	வழிப்பறி raiding travellers	நிரை மேய்த்தல் herding cattle	தேன் எடுத்தல் gathering honey	களை கட்டல் weeding	மீன் பிடித்தல் fishing
	நிரை கவர்தல் cattle raids		தினை காத்தல் guarding millet	நெல்லரிதல் reaping paddy	உப்பு விளைத்தல் making salt
ஊர் SETTLEMENT	குறும்பு fort-hamlet	பாடி herders' camp	சிறுகுடி hill hamlet	ஊர் village	பட்டினம் port town
					பாக்கம் fishing hamlet

தெய்வம் உணாவே மாமரம் புள் பறை
செய்தி யாழின் பகுதியொடு தொகைஇ
அவ்வகை பிறவும் கருவென மொழிப

Tolkāppiyam, Akattiṇaiyiyal — deity, food, beast, tree, bird, drum, occupation and the yazh: these, and things of their kind, are called karu.

CHECKED AGAINST

Tolkāppiyam, Poruḷatikāram, Akattiṇaiyiyal (the karu sūtra quoted above)

Sangam landscape — Wikipedia

Eco-regions during the Sangam age — Rau's IAS

Landscape in Sangam literature — P. Aruna Devi

How Sangam literature imagined a coastal world — The Wire

Also exported as data: `import { KARUPPORUL } from "@/lib/sangam/karupporul"`

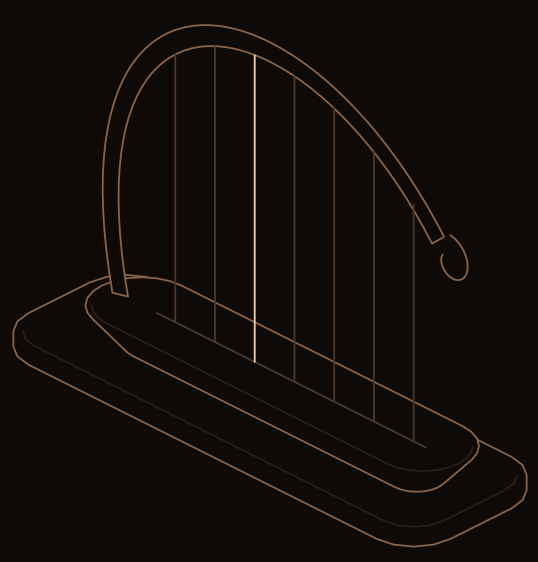
Sangam objects · பொருள்கள்

Things from the Sangam world drawn as Hairline objects, ready to sit in any landscape: they take whatever ink they are placed in. Move over each one.

யாழ் Yazh

karu · yazh

rest



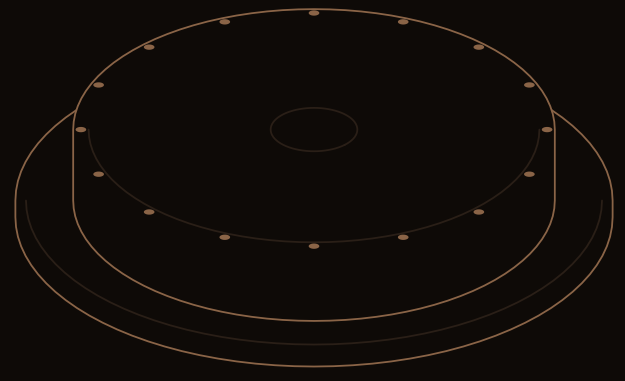
sympathy

The bow harp of the poems, and the instrument Yazhi is named for. Tolkāppiyam counts the yazh among the karu, so each landscape has its own.

பறை Parai

karu · drum

rest

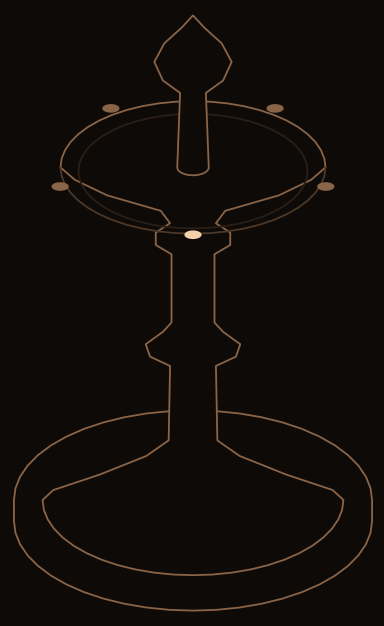


ring speed

The frame drum. Each tinai has its own parai in the karu, beaten for its own occasions — harvest, the catch, the bull-taming.

விளக்கு Vilakku

rest

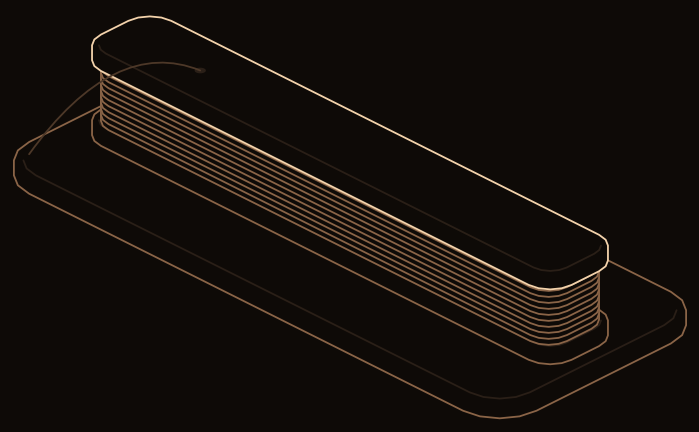


catch delay

The standing oil lamp of the Tamil home and temple: a turned brass solid with wicks round its bowl.

ஓலைச்சுவடி Olai chuvadi

rest



fan spread

A palm-leaf manuscript. The Sangam anthologies came down to us on leaves like these, copied by hand until they were printed in the nineteenth century.

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Foundations · அடித்தளம்

Small set, strictly used. Everything on this page is built from these and nothing else.

Tokens

`--env-ground`	Page ground. Five near-blacks, each tinted toward its landscape.
`--env-surface`	Raised surfaces: panels, composer, the tray. One step up from ground; never a shadow.
`--env-hi`	The one bright stroke. One per view at rest: the active bubble, the chosen part.
`--env-edge`	Silhouettes and rules. The .hair and .hair-lo borders mix it down.
`--env-mid / --env-lo`	Ordinary and receding strokes, for depth without fills.
`--env-accent`	The tinai hue, also exposed as --accent for the existing library.
`--env-text / --env-text-dim`	Cream text, warmed or cooled toward the landscape.
`--hairline-*`	Mapped onto the same values so Hairline figures sit in the landscape's ink.

Type

Display, for names and moments

யாமறிந்த மொழிகளிலே

Body, for reading. Latin leading 1.55; Tamil 1.75 in text and 1.35 in display, because kombu and the vowel signs need the room.

`Mono, for tokens, read-outs and numbers · 0123456789`

Line and shape



Bubbles come in three sizes, 1 : 0.74 : 0.55, sized by how much work lives behind them. One stroke weight everywhere: a 1px hairline in the landscape's edge colour. No drop shadows, no glows — depth comes from ground, surface and stroke.

Motion · இயக்கம்

Two clocks, and a landscape to set their character.

A tween for choices

Anything you choose — a tab, a bubble, a step — moves on one 700ms curve, cubic-bezier(.32,.72,0,1). Calm, decisive, the same everywhere.

Springs for the pointer

Anything that follows the hand rides a spring, and the spring belongs to the tinai: heavy in Palai, buoyant in Kurinji, exact in Marutham, loose in Neytal.

Stagger by distance

When one change spreads, it starts where you acted and reaches neighbours in order of distance, 30–60ms apart. Try the tray on the cover.



With reduced motion, springs and tweens jump to their end, ambient fields draw one still frame and colour changes are instant. Nothing is lost: every state is still reachable and labelled.

Access and Tamil · அணுகல்

Hit areas stay put

Only the drawing moves. A bubble that swells keeps its button exactly where it was, so the pointer never chases a target.

Everything has a name

Figures are role=img with a sentence of meaning; their read-out is a polite live region. Ambient canvases are aria-hidden.

Keyboard is a pointer

Focus lifts a bubble the way hover does. Focus rings use the bright stroke, never colour alone.

Tamil, whole

Never letter-spaced. Text is cut by grapheme (Intl.Segmenter), so கு is never torn into க and ு.

Contrast

Text and the bright stroke stay above WCAG AA on every ground; accents carry meaning only alongside a word.

Sleep

Every animated surface sleeps when offscreen or when the tab is hidden. Batteries are a resource of the person, not of the app.

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Do and don’t · செய் · செய்யாதே

DO Pick the tinai from what the app is for.	DON'T Pick a tinai because you like its colour.
DO One bright mark at rest.	DON'T Light up everything that is interactive.
DO Let the landscape move the ambient field.	DON'T Put text or controls on the ambient field.
DO Mark a draft as a draft.	DON'T Ship Tamil copy no native speaker has read as final.
DO Change landscape across whole sections.	DON'T Mix two tinai inside one card.

Use it

```
import { TinaiScope, TinaiAmbient,
  MagneticTray, HairlineFigure,
  TINAI_FIGURES } from "@/ui";
```

Library

Components

Modules

Pages

Credits

Hairline kernel and rules © Lucas Marques, MIT licence, vendored in `src/ui/hairline`. Tinai moods from the Sangam landscape tradition as written up on yazhi.dev.